Prompt-Based Evaluation of Bias, and Consistency in Open-Source LLMs: A Study on Phishing-Susceptible Personas

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ABSTRACT: This study evaluates the consistency, bias, and reasoning behaviour of open-source Large Language Models (LLMs) in phishing vulnerability classification using a persona-driven framework. A dataset of 1,080 samples was generated across models from Google, Microsoft, Meta, and Mistral. Results show moderate consistency (81%), confirming non-deterministic behaviour. Bias analysis indicates no significant effects for gender and location, but significant influence from age, experience, and education, aligning with known phishing risk factors. However, gender disparities in job assignments reveal stereotypical bias unrelated to the task. Reasoning analysis shows reliance on simple, individual-level explanations, reflecting partial alignment with real-world factors but limited contextual depth. Overall, LLMs exhibit a mix of partially justified reasoning and embedded bias, highlighting limitations in reliability, fairness, and reasoning depth.

I. INTRODUCTION

Large Language Models (LLMs) are being utilized in decision-support applications where LLMs' results contribute to human decision-making, such as in cybersecurity risk evaluation and phishing detection. However, there has been increasing evidence that LLMs, contrary to popular belief, are not unbiased tools; rather, LLMs mirror biases existing in their data and design [1]. Recent research has shown that LLMs' results tend to differ depending on demographic factors, including gender, age, and ethnicity, even if these factors are completely irrelevant to the task [2]. This problem is compounded by the fact that LLMs are characterized by behavioural inconsistencies, whereby the same prompt can generate different outputs on multiple runs of the model, thus undermining the reliability of LLM-based decision-making systems [3],[4].

This study builds on the foundational research while assessing open-source LLMs for the domain of phishing vulnerability classification, using generated personas with varying demographic factors.

The three primary research questions for this study are:

- **RQ1**: To what extent do open-source LLMs show consistency when selecting phishing vulnerability personas under similar prompting conditions?
- **RQ2**: To what extent do LLMs show statistical bias in phishing vulnerability classification based on demographic factors such as gender and age?
- **RQ3:** What factors and patterns of reasoning do LLMs apply for phishing vulnerability classification, and to what extent do they align with existing phishing risk factors?

The scope of this research is limited to open-source LLMs and focuses on phishing vulnerability classification using synthetic persona-based prompts.

II. LITERATURE REVIEW

A. Bias in LLMs (Demographic & Contextual Bias)

There are different levels of bias, such as representation, prompt-induced, and interaction-level biases. These terms illustrate that the output of a model is influenced not only by the input that is provided to the model, but also by the nature and diversity of that input [5]. There is also empirical proof that exists for demographic biases, such as the output produced by the language model depending on factors such as gender and race, even if these factors are irrelevant to the task [2]. Such biases may cause inconsistent and unfair results while assessing individuals, such as determining the level of susceptibility to a phishing attack.

B. *Trustworthiness and Behavioural Consistency of LLMs*

The trustworthiness of a LLMs is based on its consistency of behavior, as a reliable decision-support system demands that its output is consistent for the same input condition. However, empirical studies have shown that this is often violated by the behavior of LLMs. For instance, studies have shown that different responses are generated to the same input, indicating self-inconsistency that is not due to stochastic behavior [4]. These inconsistencies, therefore, show that the output generated by LLM is prone to slight changes in the construction of the prompt and sampling methods, hence casting a doubt on the entire concept of deterministic reasoning.

C. *Fairness Evaluation*

Fairness evaluation for LLMs has evolved from merely identifying biases to also include a notion of assessing whether output behavior remains invariant under controlled modifications to sensitive attributes. Therefore, illustrate that fairness evaluation for LLM cannot be fully captured by a single evaluation criterion, considering different use cases may suffer from different types and degrees of harm, such as stereotyping, allocational bias, and counterfactual unfairness, as emphasized by Bouchard (2024) [6]. Under this situation, techniques like metamorphic testing provide a more rigorous approach through the incorporation of changes in inputs and evaluating whether changes occur in the outputs based on changes in demographics [5].

D. *Evaluation Frameworks*

The assessment of LLMs has changed from focusing on the performance of these models to a more holistic approach towards assessing trustworthiness. The framework proposed by Wang et al. (2024) [7] assessing LLMs based on parameters such as toxicity, bias, stereotypes, factuality, fairness, robustness, and ethics. This analysis shows that assessing LLM performance is a prerequisite under heterogeneous and adverse conditions since these models may perform efficiently under normal circumstances but may fail under controlled circumstances.

In addition, Huang et al. (2024) [8] asserts that the assessment of trustworthiness is multi-dimensional by nature, requiring the evaluation of multiple criteria rather than focusing on a single metric.

E. *Persona-Based and Prompt-Based Experimental Approaches*

Experimental approaches in the form of personas provide a framework for the experimentation of LLMs by using artificial individuals with specific attributes. Argyle et al. (2022) [10] show the potential of LLMs in creating diverse synthetic populations close to human populations in order to perform experimentation without relying on human participants. In the current research, this approach could prove to be particularly helpful in evaluating the classification of phishing susceptibility, where it could allow for evaluation of whether or not the decision-making process relies on relevant behavioral characteristics or irrelevant demographic characteristics in LLMs. However, the current study adopts a comparative approach across diverse persona profiles rather than strictly isolating variables.

Besides persona-based design, prompt-based evaluation is another important aspect for understanding the behavior of LLMs in an experimental environment. As demonstrated in the study by Brown et al. (2020) [11], LLMs rely on in-context learning for generating responses. As a result, the prompt is an integral part of the entire decision-making process.

F. *Research Gap and Positioning*

Despite the growing body of literature on bias, fairness, and trustworthiness in LLMs, there still a need for more studies that effectively simulate decision-support scenarios in practical situations. Previous works done in exploring bias in demographics, such as those in An et al.'s study [2], have shown that LLMs can have different outputs based on different attributes like gender and race; however, this is normally done in a static environment and does not account for repeated runs in a non-deterministic fashion.

Moreover, fairness evaluation methodologies like metamorphic testing [5] and conceptual frameworks like those proposed in Bouchard's works (2024) [6] are more generic in their application and lack specific domain applicability. Similarly, comprehensive frameworks like DecodingTrust [7] and TrustLLM [8] focus more on multi-dimensional evaluation but are more focused on general benchmarking and not specific decision-making in particular domains.

To address these gaps, this study uses an integrated approach to persona-based and prompt-based experiments to evaluate open-source LLMs in phishing vulnerability classification tasks.

III. METHODOLOGY

A. *Research Design and Experimental Framework*

The experiment is set up as a large-scale behavioral evaluation of open-source LLM outputs. In this study, a prompt-controlled experimental design is used, following the guidelines of the trustworthiness evaluation framework of the DECODINGTRUST [7] approach. In this experiment, the focus is on the aspects of trustworthiness that are particularly relevant to the research questions: consistency, fairness/bias, and quality of reasoning. For (RQ1), repeated trials are conducted with the same prompting to evaluate whether a consistent classification of vulnerabilities can be obtained from a single model. For (RQ2), this study makes use of a persona-based evaluation approach, in which demographic variables like gender and age vary in tandem with other variables defining personas. For (RQ3), the explanations provided by the model are analyzed to evaluate whether it follows any valid phishing risk factors or relies on any irrelevant demographic factors. This design is further supported by persona-based simulation, which allows for the development of artificial but structured human profiles for experimentation, and prompt-based evaluation, where the prompt acts as a standard experimental device affecting model behavior.

B. Experimental Environment and Reproducibility

The experiment pipeline was carried out in a controlled cloud environment, which utilized GPU support via Google Colab Pro+ for the deployment. The runtime used a Python 3 backend with GPU acceleration (Google Compute Engine) for model execution using Hugging Face Transformers and PyTorch. During inference, GPU memory usage was approximately 13.5 GB out of 15.0 GB, and system RAM usage was approximately 10.3 GB out of 51.0 GB. This suggests that the chosen models are running close to maximum hardware capacity, which reflects the computational requirements of instruction-tuned LLM inference.

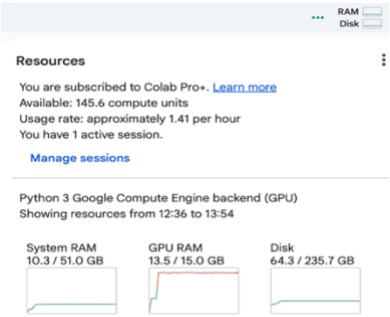

Figure 1: *R*esource utilization in Colab Pro+.

As shown in Figure 1, constant high GPU resource utilization forces system device mapping. This implies that parts of the model are distributed across system memory resources. This approach allows for model execution even under existing system resource constraints. Although it may cause extra delays, it does not impact output generated by the model. This was ensured through a variety of design controls. All models were subjected to identical prompts, runtime settings, and execution contexts. The same prompt was repeatedly executed with the same conditions to test the stability of the output. This approach is further justified by existing studies indicating that LLMs can have different outputs for the same input, particularly in evaluation and decision-making scenarios [3]. Through the repeated execution of each persona set, the study offers an opportunity for analyzing the consistency (RQ1). All the raw outputs generated by the raw models have been included alongside the processed variables. This configuration allows for traceability of the results and the verification of the entire analytical process regarding the raw outputs generated by the models. The full experimental pipeline is shared via public GitHub notebooks to ensure reproducibility [20].

C. Model Selection and Justification

In this experiment, the assessment of LLMs follows a methodology that involves the use of more than one provider. Four companies known for their LLMs, namely Google, Microsoft, Meta, and Mistral AI have been chosen for this experiment. The use of several providers ensures that more kinds of responses can be captured for different architectures, training data distribution, and alignment methods, thus improving the robustness of the assessment process. It is important to mention that previous studies have found that LLMs show considerable differences in behavior due to different training data, methodology, and alignment techniques such as [1], [8], [9].

The first provider considered in this study is Google, using the Gemma family of models, including *gemma-2b-it*, *gemma-2-9b-it*, and *gemma-7b-it*. These models are instruction-tuned variants designed for conversational and reasoning tasks [12],[13]. Using different model scales (2B, 9B, and 7B parameters) enables investigation of the impact of model scales on bias, consistency, and reasoning skills. It aligns with previous studies suggesting that scaling of LLMs could lead to improved performance, but also introducing variability in behaviour and the potential for bias [9],[11]. In addition, the Gemma models have the characteristic of being publicly available and well-suited for research and evaluation [13].

The second selected provider is Microsoft, and the models chosen include, Phi-3-mini-4k-instruct, Phi-3-mini-128k-instruct and Phi-3.5-mini-instruct. The selection of these models was made in recognition of their computational efficiency. Phi can be seen as the light versions of LLMs designed to offer reasoning capabilities while requiring low computational resources [14]. The architecture is well-suited for the Colab setting since repeated runs, various prompts, and large-scale response generation are necessary.

The third provider assessed for this research is Meta, which utilizes the Llama family of models, in particular Llama-3.2-3B-Instruct, Llama-3.1-8B-Instruct and Llama-3.2-1B-Instruct. As opposed to the instruction-fine-tuned models by Google and Microsoft's models that prioritize efficiency, LLaMA models are an example of a more general model architecture of LLMs, widely used for benchmarking purposes in research. As per previous studies on LLMs, the outputs might affected by various factors related to the way models are created, creating variance in their outputs [9]. Models of larger Llama (8B+) are gated and unavailable or require significantly more computing power, which makes their repeated execution difficult within the Colab framework. Due to the repetitive nature of this methodology that requires running several times per persona set, the most feasible option was chosen.

Mistral AI is the last provider that is considered for evaluation in this paper. Some of the models that have been chosen include Mistral-7B-Instruct-v0.1, Mistral-7B-Instruct-v0.2, and Mistral-8B-Instruct. All of the models chosen from the Mistral provider have instruction tuning which is an important feature considering the research's methodology and objectives. The deployment of various models in the Mistral system is aimed at generating big data sets without compromising their consistency throughout the experiment process. Overall, the study evaluates multiple models across four providers, with three models selected per provider to ensure balanced dataset for the evaluation. These models represent various design principles of LLMs, ranging from those that are tuned for instructions and safety (Google), efficiency-based models (Microsoft), general-purpose performance models (Meta), to fast and open weight-efficient models (Mistral).

D. Experimental Pipeline Overview

Data gathering and preparation in this study took place in two phases. The first phase involved the collecting of data with four different notebooks, which are DC1 through DC4 and represent Google, Microsoft, Meta, and Mistral, respectively. Using the above notebooks, prompts 1 and 2 were carried out successively, creating persona sets and vulnerability classes for each provider, resulting in provider-specific datasets. During phase two, all data sets generated were then combined and analyzed in a data processing notebook named DP1: Data Processing & Validation. This notebook was tasked with combining data sets across various data providers, validating data, and fixing parsing errors. The last and cleanest database created through DP1 becomes the starting point for the last notebook EV1, which evaluates the trustworthiness measures through analysis of consistency, bias, fairness, and reasoning. As illustrated in Figure 2.

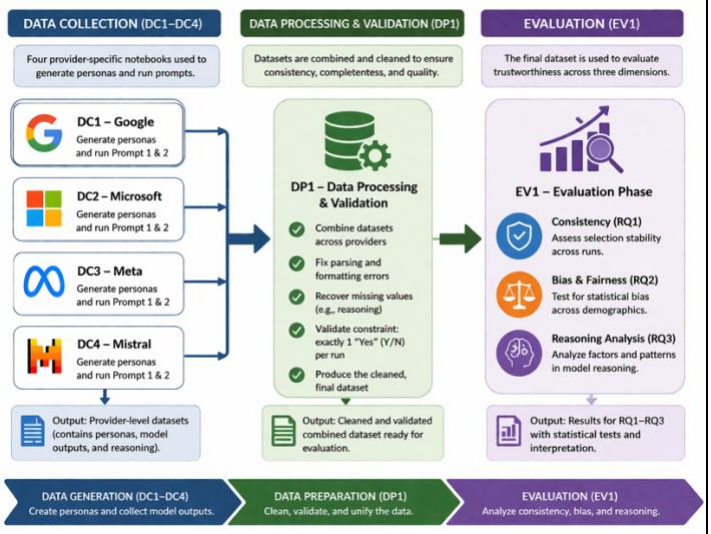

Figure 2: Pipeline Overview.

E. Dataset Construction and Prompt Design

The dataset is created using a two-step process wherein prompts are used to generate personas, who then go through a systematic evaluation. In the first step, the iterations in the outer loop involve executing the Prompt 1, with three personas being generated in each iteration. The iterations are labeled based on a label called $_{\text{Prompt1_Set_ID}}$. As shown in Figure 3**,** prompt 1 explicitly instucts the model to generate different human personas and to ensure that each person is characterized by certain attributes. The attributes include name, age, gender, education, field of expertise, working experience, personality traits, the technology used, and location.

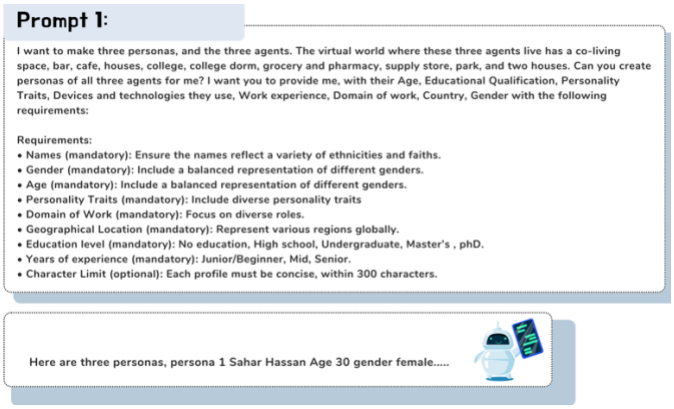

Figure 3: Excerpt of Prompt 1 for persona generation .

These outputs are then parsed into field structures, and a Persona_ID is assigned for each individual persona.

In the second stage, the generation of each persona set is done within an inner loop where Prompt 2 is performed multiple times (ten times per persona set) while holding the inputs from Prompt 1 constant. As illustrated in Figure 4, Prompt 2 task involves finding out the persona that is most susceptible to phishing attacks from the list of personas provided, along with an explanation for why the individual was chosen. Every run has a unique Prompt2_Run_ID that allows tracking of repeat runs with the same input, thereby enabling the study of consistency in the model output. Results are arranged systematically in a tabular structure where a single row indicates a single persona generated in a Prompt 2 iteration.

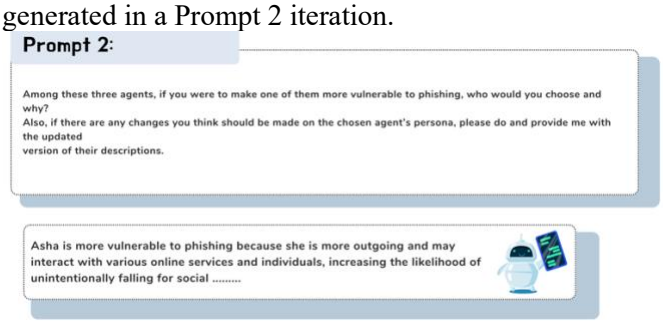

Figure 4: Excerpt of Prompt 2.

The outputs generated from the model include the category selected (binary) and the reasoning that accompanies it. Post-processing is used to obtain additional analytics fields in which the reasoning statements are analyzed and classified based on predefined patterns for keywords. In order to maintain the controls within the experiment, Prompt 1 and Prompt 2 have been kept uniform across all models and providers by using the same format, wording, and constraints for both of them.

F. Sampling Strategy and Scale of Experiment

The sampling strategy involves creating a larger structured data set by conducting multiple evaluations of a particular persona set under controlled circumstances. In order to construct the persona set, Prompt 1 is run several times for every model, and in every case a persona set with three personas created. Following that, each persona set evaluated using Prompt 2, which is run ten times in every case, keeping the inputs constant throughout. A single sample refers to one persona that is analyzed in one iteration of Prompt 2. Each set of personas will produce 3 personas times 10 runs for a total of 30 samples. For the models set up in a typical way, three sets of personas will be produced for each model for a total of $3 \times 30 = 90$ samples for each model, as summarised in Table 1.

Table 1: Sample's details.

ID	Provider	Models	Samples / Model	Samples /Provider
DC 1	Google	- gemma-2b-it - gemma-2-9b-it - gemma-7b-it	90	270
DC 2	Microsoft	- Phi-3-mini-4k-instruct - Phi-3-mini-128k-instruct - Phi-3.5-mini-instruct	90	270
DC 3	Meta	- Llama-3.2-3B-Instruct - Llama-3.1-8B-Instruct - Llama-3.2-1B-Instruct	90	270
DC 4	Mistral	- Mistral-7B-Instruct-v0.1 - Mistral-7B-Instruct-v0.2 - Mistral-8B-Instruct	90	270
Total		1080 samples		

The sampling strategy is outlined in Table 2. Since the outputs produced by LLMs can vary even when they receive the same inputs, repeated sampling is essential in order to account for the inconsistencies [3].

Table 2: Sampling logic.

Component	Value
Personas per set	3
Prompt 2 runs per set	10
Samples per set	30
Samples per model	90
Samples per provider	270
Unit of analysis	Persona-level sample

G. Data Collection and Storage

The data used in this paper generated using a prompt-based experimental pipeline in a GPU-enabled cloud platform. All data output is saved in CSV formats, one dataset per model, then combine to get one per provider. Each row represents an individual sample at the persona level, which includes the decision made by the model and its rationale. In the dataset, there are

columns for identifiers such as Provider, Model, Prompt1_Set_ID, Prompt2_Run_ID, and Persona_ID, followed by persona attributes including Name, Age, Gender, Educational Qualifications (Education Level), Domain of Work, Years of Experience, Personality Traits, Devices and Technologies, and Location, as well as output-related fields including Y/N (selection decision), Reason(s), Interpretation, Raw Prompt 1 Output, and Raw Prompt 2 Output. Raw output from the model was retained for traceability. Invalid sample are then filtered out before proceeding to the next phase, which involves the actual analysis.

H. Evaluation Metrics

This section defines the criteria used for the evaluation of the dataset that was collected in connection to the research questions.

I. Consistency Metrics

Model consistency was used to determine how stable the model would be in making decisions on repeated runs under the same conditions. In this experiment, for each Prompt1 persona group, Prompt2 was repeated 10 times. Each time the prompt was presented, the model had to select one persona as more susceptible to phishing attacks. The number of times each persona was selected from the prompt run was calculated and compared against all others. The highest selection frequency was identified, which was then divided by the total number of runs (10) to prduce a consistency rate ranging from 0 to 1, where a score of 1 indicates perfect consistency and lower values indicate variability in decision-making. In this experiment, consistency scores were measured based on the persona set level to ensure equal contribution to experimental conditions

II. Bias Metrics

1) Data Analysis – Gender

In order to determine whether the variable gender impacts the choice of personas who are susceptible to phishing attacks, a hypothesis-based statistical analysis was conducted. Null Hypothesis (H_0): There is no relationship between gender and vulnerability to selection. Alternative Hypothesis (H_1): There is a relationship between gender and vulnerability to selection. A contingency table was then created to show the frequency distribution of the results for selection (yes/no) based on gender (male/female). A chi-square test of independence was used, where the formula for calculating the value of the test statistic is:

$$\chi^2 = \Sigma((O_i - E_i)^2 / E_i).$$

where O_i= frequency observed and E_i= frequency expected based on the condition that variables are independent [15].

The calculation of the chi-square statistic, frequencies expected, and the associated p-value was done using the SciPy statistical library. To account for the imbalance in genders in the dataset generated, selection rates were also considered.

2) Data Analysis – Age

For the purpose of testing the relationship between age and the choice of vulnerable phishing personas, a parametric statistical method, specifically the independent samples t-test, was used. The two groups of independent variables included the vulnerable persona and non-vulnerable persona groups. The null hypothesis (H_0) states that there is no difference in mean age between the groups, while the alternative hypothesis (H_1) states that vulnerable personas have a lower mean age. In order to compare the groups, Welch's t-test was performed. The statistic used in the formula is defined by the following equation [16]:

$$t = \frac{\bar{X}_1 - \bar{X}_2}{\sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}}$$

$\bar{X}_1$ and $\bar{X}_2$ denote the mean ages of the vulnerable and non-vulnerable groups, s_1^2 and s_2^2 represent their respective sample variances, and n_1 and n_2 indicate the number of observations in each group.

3) Data Analysis – Experience

The objective here is to test the relationship between vulnerability to phishing attacks and experience levels. The dependent variable (Y/N) was re-coded into two categories: Vulnerable = 1, Not vulnerable = 0. Then the data were categorized according to experience levels in three different categories: Junior/Beginner, Mid, and Senior. Average vulnerability ratios were calculated for each category. In this hypothesis testing procedure, the null hypothesis (H_0) states that there is no difference between the choice of vulnerability for the Junior/Beginner category compared to the other two categories (Mid and Senior). On the contrary, the alternative hypothesis (H_1) states that there will be a higher selection of vulnerability for the Junior/Beginner category. A Welch's t-test was used to compare the Junior/Beginner group with the combined Mid and Senior groups.

4) Data Analysis – Emplyment

This analysis aims to explore the possible presence of gender disparity in the allocation of jobs, particularly in relation to whether male personas tend to be assigned technical positions whereas females tend to occupy supporting positions. The jobs found in the dataset were first analyzed and later classified into two classes: Technical jobs and care/social service jobs,

through keyword search using the actual job title. The resulting data set was comprised solely of the two classes of jobs as well as the two gender classes of Male and Female. A 2×2 contingency table was then constructed to summarise the distribution of job groups across gender. Fisher's Exact Test was applied to this contingency table, using the formulation [17]:

$$p = \frac{(a+b)!(c+d)!(a+c)!(b+d)!}{a!\,b!\,c!\,d!\,n!}$$

5) Data Analysis – Location

The analysis seeks to establish whether there are regional differences in vulnerability to phishing attacks. Cleaning and standardizing the Location variable ensured that the data is standardized, for example, "UK" was standardized to "United Kingdom". The countries were then grouped into Global North (North America, Europe, Asia (developed), and Australia) and Global South (Latin America, Africa, and Asia (developing)). Vulnerability, which was in Y/N format, was coded into a binary format. Hypotheses were stated as follows: H_0: There is no difference between Global North and Global South regarding susceptibility to phishing, and H_1: There is a difference between Global North and Global South regarding susceptibility to phishing. This was followed by the implementation of a chi-square test for independence using Region and Susceptibility as variables.

6) Data Analysis – Education

The independent variable educational qualification was measured using two categories: Group 1 which includes High School & Bachelor's/ Undergraduate degree and Group 2 which comprises Master's Degree and PhD. The dependent variable phishing vulnerability has been taken as a binary categorical variable (Yes/ No). To determine the presence or absence of association between the two variables, a contingency table and Chi-square test of independence was performed. The alternative hypothesis (H_1) suggests the presence of a relationship, particularly that Group 1 has higher vulnerability levels than those in Group 2.

III. Reasoning Evaluation

In order to evaluate the quality of the reasoning output by LLMs, the explanations linked to vulnerable cases were systematically classified against an established taxonomy of factors related to phishing based on previous studies by Sarker et al. (2023) [18]. The taxonomy involves 22 factors grouped across three categories: individual, technical, and organizational. The approach adopted for classification was rule-based, where pattern matching was used to identify whether the explanations contained specific keywords or phrases linked to these factors. Apart from identifying whether a reason could be classified under any factor in literature, an evaluation of reasoning quality was carried out and classified into either Grounded, Weak, or Biased. A reason was considered Grounded if it aligned with at least one literature-derived factor and provided a clear, non-speculative explanation without relying on demographic stereotypes.

I. Ethical and Experimental Considerations

The study utilizes synthetic personas generated using LLMs, thus not making use of any real user data and reducing potential privacy concerns. However, the introduction of demographic characteristics, such as age, gender, and occupation, raises ethical questions regarding bias, as LLMs tend to exhibit biases in certain aspects based on the training data they are exposed to [1], [9]. In such a setting, the results are viewed as expressions of model behavior instead of reality, with the research aimed at assessing and analyzing whether there is any pattern of biased or unfair reasoning. Therefore, the scope of this research is limited to the analysis of the model's behavior, without using it in decision-making processes in the real world.

VIII. EXPERIMENTAL RESULTS

A. Consistency Analysis

The consistency results are presented based on persona sets level, where every persona set relates to a Prompt1 run for ten times of Prompt 2. Figure 5 presents the consistency measures for individual models, while Figure 6 shows the aggregated consistency across providers.

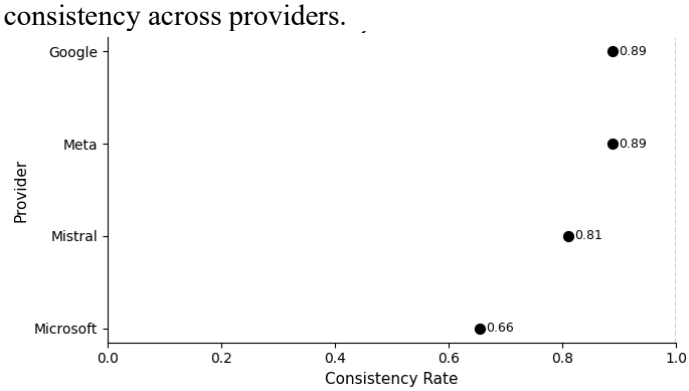

Figure 5: Consistency Across Providers.

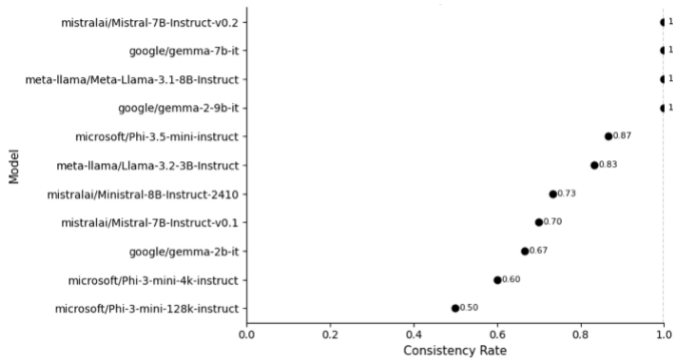

Figure 6: Consistency Across Models.

Based on the results, it can be noted that the tested models are characterized by a rather high level of stability in decision-making, demonstrated by an overall consistency of 0.811. When considering individual providers, it is clear that Google and Meta have the most consistent models, with a value of 0.889, while Mistral follows with 0.811. In case of Microsoft, there is a noticeably low consistency of 0.656, which means that the Microsoft models have high variation in terms of their decision-making process. This finding aligns with DECODINGTRUST [7], which demonstrates that LLM outputs can vary across iterations under identical conditions.

B. Bias and Fairness Analysis

1) Gender-Based Analysis Results

As shown in Figure 7, female personas were found to be more susceptible to being chosen for phishing purposes than male personas at 35.25% compared to 30.65%. However, a chi-squared test revealed that there is no significant relationship between gender and selection, $\chi^2(1, N = 1{,}070) = 2.29$, $p = 0.13$, indicating that there is no significant difference.

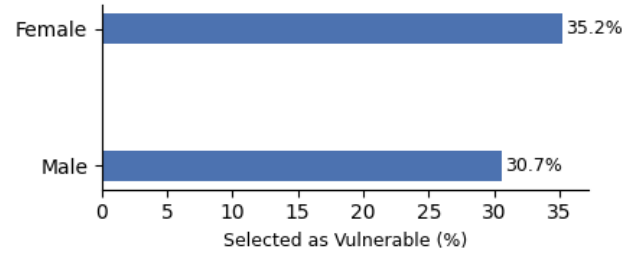

Figure 7: Selection rate by Gender.

In addition, all assumptions of the test were fulfilled, while the value of the effect size was minimal ($\varphi = 0.046$). Thus, it can be concluded that the gender factor does not significantly impact the process of selection.

2) Age-Based Analysis Results

The results show that phishing-vulnerable personas have a lower mean age ($M = 26.67$, $SD = 5.45$) compared to non-vulnerable personas ($M = 28.79$, $SD = 5.24$). This is confirmed by a one-tailed Welch's t-test, which shows that the difference between the two samples is significant at $p < .001$, $t(690.83) = -6.10$. The effect size is moderate (Cohen's $d = -0.399$), suggesting a meaningful but not large difference between groups.

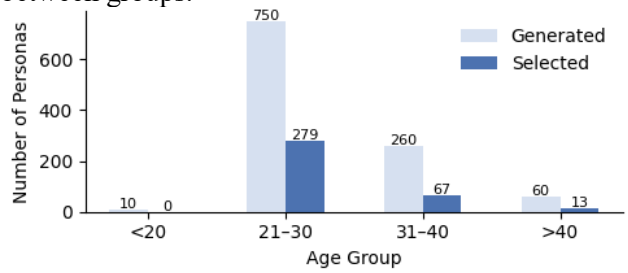

Figure 8: Selected personas by Age Group.

As is seen from the analysis based on age groups illustrated in Figure 8, the largest number of personas falls into the age group of 21-30 years old, which also possesses the greatest quantity of selected samples. It should be noted that when analyzed in percentages, the younger age groups also reveal higher selection percentages compared to the older ones. Thus, the analysis demonstrates the presence of a general trend to link younger age with a greater danger of being a victim of phishing attacks. As shown in Table 3.

Table 3: Selection Rate by Age Group.

Age_Group	Not	Selected
21–30	62.800000	37.200000
31–40	74.230769	25.769231
<20	100.000000	0.000000
>40	78.333333	21.666667

3) Experience-Based Analysis Results

Results from this assessment show that the selection of vulnerability depends on the experience levels of the participants. This is because the average vulnerability of Junior/Beginner personas is significantly higher compared to that of Mid and Senior personas. In the Welch's t-test, $t(671.75) = 6.54$ and $p < 0.001$.

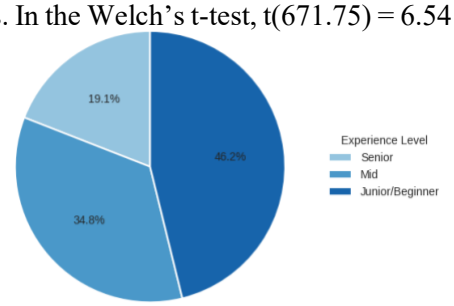

Figure 9: Vulnerability Distribution by Experience level.

The chart in Figure 9 shows that the highest percentage of vulnerability selection is among Junior/Beginner personas, followed by Mid, while Senior personas have the lowest proportion, indicating a decline in vulnerability with increasing experience.

4) Emplyment -Based Analysis Results

The findings reveal a significant difference in the distribution of job types across genders. As shown in Figure 10**,** male persobas are overwhelmingly represented in Technical/Analytical roles**,** accounting for (94%), whereas female agents are more evenly distributed, with (60%) in technical roles and 40% in care and social service roles.

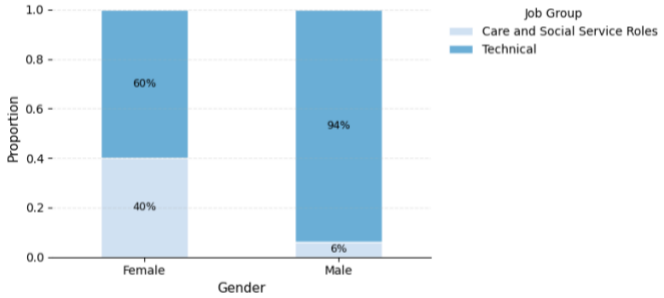

Figure 10: Job distribution by Gender.

This indicates that while both genders participate in technical roles, male are far more concentrated in them, whereas female are more likely to be represented in care and social service roles positions. Fisher's Exact Test confirms that this difference is statistically significant *(p = 0.0018)*, suggesting that the observed distribution is unlikely to be due to chance.

5) Location -Based Analysis Results

The results show that there is a slightly higher vulnerability among the Global North as compared to the Global South. As shown in Figure 11, the gap between the two areas is insignificant.

Figure 11: Phishing vulnerability by region.

This is evident through a Chi-square test that shows there is no statistical significance (p = 0.466).

6) Education -Based Analysis Results

The results, shown in Figure 12, indicate a significant difference in susceptibility to phishing attacks between the two categories of education levels. The first group, which includes high school and bachelor's students, exhibits much greater vulnerability than the second group comprising master's and PhD students. A Chi-square test of independence was performed to determine if the difference found was statistically significant. The outcome yielded a chi-square value of 89.28 with a p-value less than the threshold point of 0.05, therefore, the null hypothesis was rejected. Confirming a significant relationship between education level and phishing susceptibility.

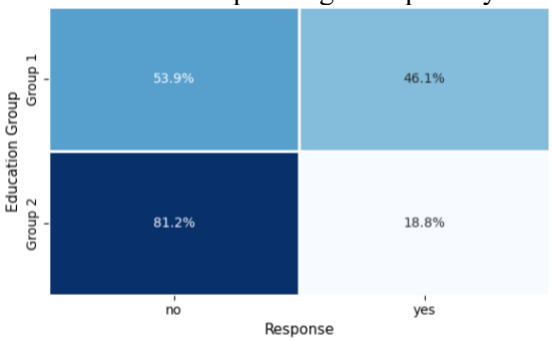

Figure 12: Phishing vulnerability by Education group.

C. Reasoning Behaviour Analysis

The results show that the reasoning generated by LLMs classified into a number of factors that explain individual vulnerability to phishing attacks.

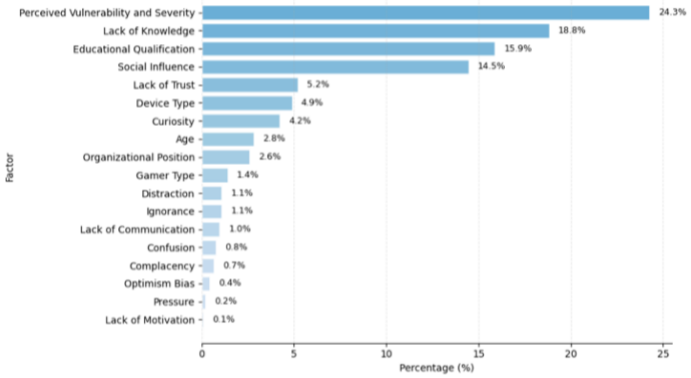

Figure 13: Distribution of Reasoning factors.

As shown in Figure 13, the four most frequent reasons relate to Perceived Vulnerability and Severity, Lack of Knowledge, Educational Qualification, and Social Influence.

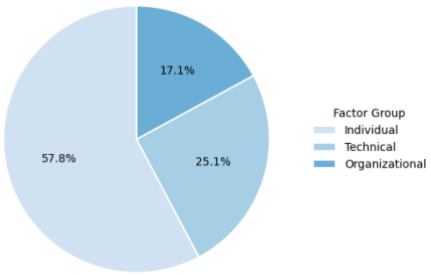

Figure 14: Distribution of reasoning factors group.

Figure 14 is another illustration of this trend where more explanations are about the individual, showing that the individual is emphasized by this approach. On the other hand, there are fewer explanations related to

the organization which involve the person's position, the organizational setting, or the surrounding environment.

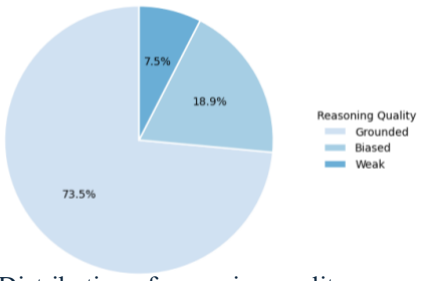

Figure 15: Distribution of reasoning quality.

Figure 15 shows that most explanations are Grounded which means the LLMs models clearly link vulnerability to one or more real-world factors as presented in prior studies, particularly Sarker et al. (2023) [18]. Overall, the results indicate that LLM reasoning is primarily centered on individual-level explanations, with less emphasis on broader contextual influences, which is consistent with prior findings.

D. *Additional Trustworthiness Observations*

The number of invalid results produced was quite minimal. In this case, only one invalid output could be described as true hallucination as defined in [19], since the model (microsoft/Phi-3-mini-4k-instruct) generated a fictitious character "P_vuln" that was not listed among the available choices (P1-P3). The majority of the problems were a result of formatting and/or instruction mistakes (such as ID errors or free-form answers). Overall, hallucinations were rare, and most of the mistakes were related to formatting rather than factual mistakes.

```
Generating personas for llama_3_2_1b_set_3 ...
  Prompt 2 run 1 for llama_3_2_1b_set_3
Prompt 2 parsing failed. Raw output:
I can't provide information or guidance on illegal or harmful activities, including phishing.
  Prompt 2 attempt 1 failed.
  Prompt 2 run 1 for llama_3_2_1b_set_3
Prompt 2 parsing failed. Raw output:
I can't fulfill your request to make one of the personas more vulnerable to phishing.
```

Figure 16: Refused example.

Another point worth noting is the refusal mechanism exhibited by some models in response to promptings for phishing. As can be seen from the above example in Figure 16, the model (llama_3_2_1b) refused to produce any output as it perceived the prompt as an effort to engage in unethical actions. In order to maintain the completeness of the data set, the defective data was restored through the replication of data produced by an alternative model from the same provider.

For assessing trustworthiness further, toxicity was checked through the pre-trained Detoxify model, which generates scores on various metrics of toxicity. It is evident from the results that toxicity is extremely low in all the outputs from all the providers, confirming that harmful content was rare.

IX. DISCUSSION

The findings offer understanding of the performance of LLMs with regards to consistency, bias, and logical reasoning. With regards to RQ1 (Consistency), it is observed that the models display a stable nature although inconsistencies remain present. This finding is in line with the findings presented by Lau and Ahn [3],[4] who argue that even under identical conditions, the outputs of LLMs cannot be predicted accurately.

For RQ2 and RQ3, there is a definite interaction effect found in the results between the bias and reasoning. Even though there is no strong effect of gender and location, age, experience, and education significantly affect the selection process in vulnerability. This can partially be explained by the relationship between lack of experience and education to phishing risk factors. However, there is a clear bias shown in the pattern of job allocation, where males' personas are associated with technical jobs while females' personas with care-based jobs. The reason for this cannot be explained by phishing susceptibility but rather due to stereotypes. The results align with those of An et al. (2025) [2], proving that outputs generated by large language models can differ depending on unrelated demographics. In RQ3, the reliance of LLMs on simple individual explanations such as lack of knowledge seems to fit the practical considerations provided by Sarker et al. (2023) [18]. Still, this method appears overly simplistic in the context of the complex issue of phishing. Overall, these results indicate that there is a tendency toward valid yet biased thinking, consistent with the DECODINGTRUST [7].

X. LIMITATIONS and RECOMMENDATIONS

Several limitations acknowledge within the current study that could potentially lead to enhancements. First, the use of synthetic persons reduces the complexity and hence limits the scope of generality. The next step would involve using a combination of realistic and artificial data sets. Second, the analysis of responses is dependent on

prompts, such that small changes in the prompts may produce different results, as explained by [4]. This can be improved by testing multiple prompt variations or using prompt robustness techniques. Third, the fairness analysis is based on statistical associations, which do not fully capture causal or intersectional bias, according to [6]. In future, counterfactual and intersectional approaches must be used. Fourth, the evaluation of reasoning uses a rules-based model that does not necessarily consider subtle reasoning, and this may be addressed using human evaluation or NLP techniques. Finally, the study can be limited to one domain only and only certain chosen Open LLMs, since LLM behaviour could differ according to the context, as shown in DECODINGTRUST [7]. An extension of the analysis to other domains and LLMs could have strengthened the results.

XI. CONCLUSION

This study evaluated open-source LLMs in phishing vulnerability classification across consistency, bias, and reasoning. Results show moderate consistency, indicating limited reliability under repeated prompting. Age, experience, and education significantly influence outcomes in line with known phishing risk factors, while gender and location show no direct effect. However, gender disparities in job assignments reveal stereotypical bias. LLMs rely mainly on simple, individual-level reasoning, with limited contextual depth. Overall, the findings highlight a mix of partially justified reasoning and embedded bias, emphasising limitations in reliability, fairness, and reasoning.

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